

D2: Scientific Calculator Gamma GUI

F4: $\Gamma(x)$ – From-Scratch Python Implementation, Version Control, and Updated Requirements

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SOEN 6011: Software Engineering Processes

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D2 scope and from-scratch boundary

D1 foundation reused

D1 established Asha as the primary user, ISO-style requirements, two candidate algorithms, and the Lanczos-with-reflection path for $\Gamma(x)$.

D2 problem coverage

- P5: from-scratch Python, Tkinter GUI, exceptions, helpful errors.
- P6: public repository, README, and high-quality commits.
- P7: complete updated requirements based on P5.

Production-code boundary

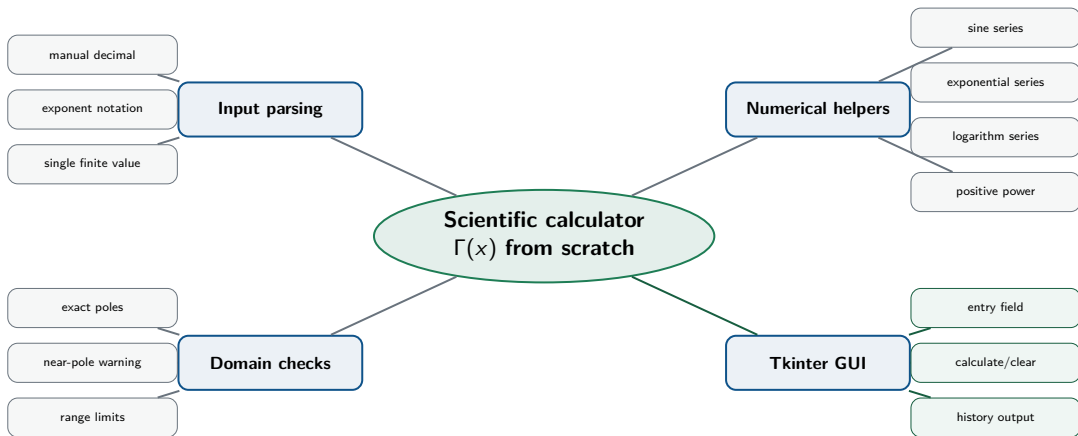
- No `math.gamma/sin/exp/log/sqrt/pow` in production.
- No `range()` in production numerical loops.
- Manual `abs_value()` and manual string counter are used.
- `math.gamma()` appears only in the validation script.

Selected algorithm retained

Lanczos approximation with Euler reflection for valid $x < 0.5$.

Sources: [1], [2], [3], [4].

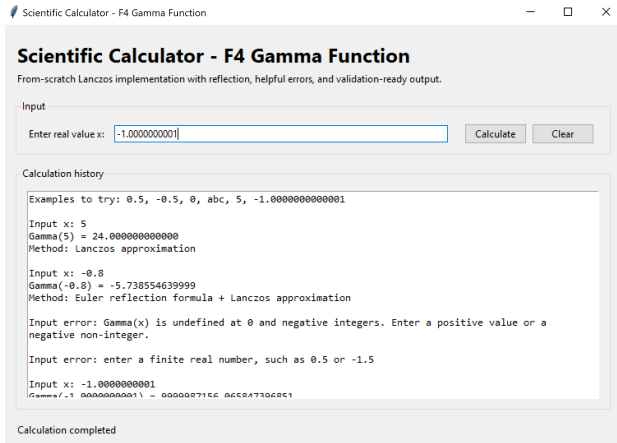
P5: Subordinate functions implemented from scratch



The D2 implementation decomposes $\Gamma(x)$ into input parsing, domain checks, subordinate numerical functions, and GUI behaviour instead of delegating numerical work to Python's math library.

Sources: [2], [4].

P5: Tkinter GUI demonstration



The screenshot demonstrates the Tkinter input field, Calculate/Clear controls, calculation history, repeated use, and helpful messages for valid values, poles, invalid text, and near-pole values.

Source: GUI execution screenshot.

P5: Four-file source structure and exceptions

Repository source files

<code>main.py</code>	Starts the GUI without an IDE.
<code>gui.py</code>	Tkinter widgets, events, and history.
<code>gamma_core.py</code>	Parsing, exceptions, numerical helpers, and $\Gamma(x)$.
<code>validate_gamma.py</code>	Validation-only reference script.

Exception handling

- `InputFormatError`
- `GammaDomainError`
- `GammaRangeError`
- `GammaUnderflowError`

Expected errors are shown in the GUI without Python tracebacks.

Built-in review

`range()` and `len()` are avoided in production numerical calculation; `math` is used only by `validate_gamma.py`.

Sources: [2], [5].

P5: GUI behaviour and numerical validation

Case	GUI behaviour
Valid input	Shows x , $\Gamma(x)$, and method path.
Exact pole	Explains why $0, -1, -2, \dots$ are invalid.
Invalid text	Asks for one finite real number.
Near pole	Warns that the result is numerically sensitive.
Overflow/underflow	Shows range-related message instead of crashing.

Validation policy

`math.gamma()` is used only in a separate validation script, not in the submitted Gamma application.

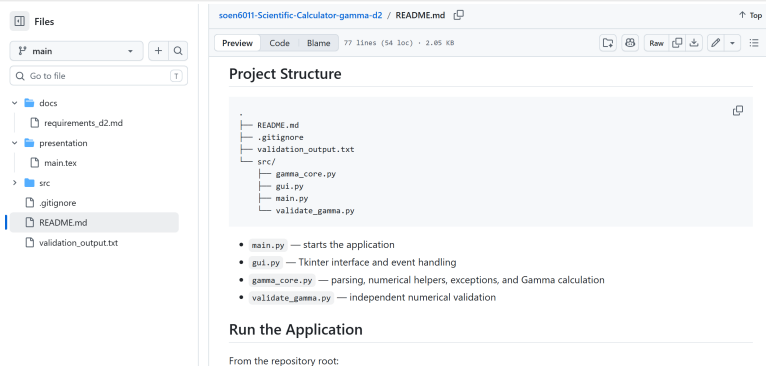
Cases	8
Max relative error	1.332×10^{-15}
Required threshold	10^{-10}
Result	Passed

Sources: [2], [3], [5].

P6: Public repository URL and contents evidence

Public repository URL

<https://github.com/arifulHossain1/soen6011-Scientific-Calculator-gamma-d2>



The repository shows the required README, validation output, and the four source files: `main.py`, `gui.py`, `gamma_core.py`, and `validate_gamma.py`.

Sources: [2], [8]; actual GitHub repository screenshot.

P6: High-quality commit history evidence

Commits on Jul 29, 2026		
Add initial Beamer presentation for Scientific calculator Gamma GUI project arifulhossain authored 10 hours ago	Verified 3723e0e	 <>
docs: add complete D2 requirements and traceability arifulhossain authored 10 hours ago	Verified 900049e	 <>
Commits on Jul 28, 2026		
refactor: remove nonessential built-ins from production numerics arifulhossain authored 11 hours ago	Verified d3a12d8	 <>
Reorder Python files in README structure arifulhossain authored 15 hours ago	Verified 5550335	 <>
docs: add project overview and execution instructions arifulhossain authored 15 hours ago	Verified 7c33e2c	 <>
docs: record numerical validation results arifulhossain authored 16 hours ago	Verified b37f218	 <>
test: add independent Gamma accuracy validation arifulhossain authored 16 hours ago	Verified ff9321c	 <>
feat: add IDE-independent application entry point arifulhossain authored 16 hours ago	Verified a53160c	 <>
feat: add Tkinter interface and calculation history arifulhossain authored 16 hours ago	Verified d8900cd	 <>
feat: add from-scratch Gamma computation core arifulhossain authored 16 hours ago	Verified 164b8bd	 <>
chore: adding ignore rules for generated project files arifulhossain authored 16 hours ago	Verified df693f0	 <>
Commits on Jul 27, 2026		

The commit history records meaningful changes for the core calculation, GUI, IDE-independent entry point, validation, and documentation instead of a single generic upload.

Sources: [2], [8]; actual GitHub commit-history screenshot.

P7: Updated requirements summary

ID	Status	Updated direction	Verification
GG-UI-01	Modified	Replace textual prompt with Tkinter entry field and buttons.	GUI inspection
GG-IN-01/02	Modified	Manual parsing accepts one finite real number and rejects invalid text.	Input tests
GG-FN-01	Modified	Compute finite real non-pole results using from-scratch Lanczos path.	Reference tests
GG-OU-01/02	Modified	Display result and method in GUI history.	Output test
GG-SE-01	Modified	Allow repeated calculations through the GUI without restarting.	Session test
GG-PR-01	Unchanged	Preserve 10^{-10} validation threshold.	Validation script
GG-SC-01/02	New	Implement subordinate numerical helpers; do not call prohibited math-library routines.	Code inspection
GG-EX-01	New	Handle expected errors without traceback.	Exception test
GG-DEP-01	New	Run from terminal without depending on a specific IDE.	Terminal test

P7 boundary

Problem 7 updates the function and GUI requirements based on Problem 5. Repository, README, and commit-message evidence remain under Problem 6 as project constraints.

Sources: [1], [2].

Appendix: P5 – CASTROFF Prompt Log

Tool used: ChatGPT.

Prompt type: Role-based, instructional, constraint-driven, iterative debugging.

Purpose: Support from-scratch Gamma implementation with Tkinter GUI.

Constraints	Avoid production use of math-library <code>Gamma/sin/exp/log/sqrt/pow</code> ; use Tkinter; provide helpful exceptions; review built-ins.
Audience	SOEN 6011 professor, TA, and students.
Structure	Boundary, subordinate functions, GUI behaviour, validation evidence.
Tone	Academic, clear, implementation-ready.
Role	Numerical-software developer and reviewer.
Output	Runnable Python source plus Beamer-ready explanation.
Focus	Gamma Guard GUI and from-scratch subordinate operations.
Function	Produce D2 P5 implementation and demonstration explanation.

Initial prompt used

Modify my D1 Gamma Guard implementation for D2. It must implement Gamma from scratch in Python, use Tkinter for a GUI, avoid prohibited math-library functions in production, and provide helpful exception handling.

Output explanation

I accepted the modular GUI structure, then modified the implementation and slides to separate validation-only `math.gamma()`, review built-ins, and document expected errors.

Source: [6].

Appendix: P6 – CASTROFF Prompt Log

Tool used: ChatGPT.

Prompt type: Instructional, planning, documentation prompt.

Purpose: Create a repository plan, README, and high-quality commit-message examples.

Constraints	Public repository, README, high-quality commits, repository URL, and screenshots of real evidence.
Audience	Marker reviewing D2 repository evidence.
Structure	Repository tree, commit sequence, README sections, evidence slides.
Tone	Practical and submission-ready.
Role	Version-control documentation assistant.
Output	README, .gitignore, commit plan, checklist, evidence slides.
Focus	Traceable Git history and clear documentation.
Function	Satisfy D2 P6 evidence requirements.

Initial prompt used

Based on the D2 project description, tell me what to upload to GitHub, how to structure the repository, and what high-quality commit messages to use.

Output explanation

I kept the repository structure and commit-message guidance, but replaced planned repository claims with screenshots from the actual public repository and commit history.

Source: [6].

Appendix: P7 – CASTROFF Prompt Log

Tool used: ChatGPT.

Prompt type: Requirements-analysis, traceability, refinement prompt.

Purpose: Update D1 requirements based on the D2 GUI and from-scratch implementation.

Constraints	Preserve D1 baseline; classify requirements as unchanged, modified, or new; keep repository items under P6.
Audience	SOEN 6011 evaluator.
Structure	Requirement ID, status, requirement direction, verification.
Tone	Formal and verifiable.
Role	Requirements engineer.
Output	Complete updated requirement list for D2.
Focus	Changes caused by D2 Problem 5.
Function	Satisfy D2 P7.

Initial prompt used

Modify my D1 requirements for D2. The implementation now has a Tkinter GUI, must be from scratch, must handle exceptions, and must have helpful messages.

Output explanation

I used the output to produce a complete P7 requirement update, removed repository requirements from the function-requirement list, and labelled each item as unchanged, modified, or new.

Source: [6].

Backup: Complete P7 requirements – input and domain

ID	Status	Updated requirement	Verification
GG-UI-01	Modified	The software shall present a Tkinter input field for one real Gamma-function argument x .	GUI inspection
GG-IN-01	Modified	When the entered text represents one finite real number, the software shall parse it using the project-defined parser.	Input test
GG-IN-02	Modified	When the entered text is not one finite real number, the GUI shall display a corrective input message.	Invalid-input test
GG-DM-01	Unchanged	When x is 0 or a negative integer, the software shall display a Gamma-domain pole message.	Boundary test
GG-DM-02	Modified	When x is within 10^{-12} of a nonpositive integer but is not exactly a pole, the GUI shall display a near-pole warning.	Sensitivity test
GG-FN-01	Modified	When x is finite, not a pole, and the result is finite in double precision, the software shall compute $\Gamma(x)$ from scratch using the selected algorithm.	Reference test
GG-RG-01	Modified	When the Gamma value exceeds the supported range, the GUI shall display a range message.	Range test
GG-RG-02	Modified	When a nonzero Gamma value cannot be represented distinctly from zero, the GUI shall display an underflow message.	Underflow test

Sources: [1], [2].

Backup: Complete P7 requirements – output and quality

ID	Status	Updated requirement	Verification
GG-OU-01	Modified	After a successful calculation, the GUI shall display x and $\Gamma(x)$ with at least ten significant digits in the history area.	Output test
GG-OU-02	Modified	After a successful calculation, the GUI shall display the algorithm path used.	Output test
GG-SE-01	Modified	After each result or error, the GUI shall permit another calculation without restarting the application.	Session test
GG-PR-01	Unchanged	For the validation set $\{0.1, 0.5, 1, 1.5, 2.5, 5, -0.5, -1.5\}$, relative error shall be no greater than 10^{-10} .	Validation script
GG-US-01	Modified	For invalid input, domain rejection, range, or underflow, the GUI shall display a plain-language message with cause and corrective direction.	Usability test

Validation-only distinction

`math.gamma()` may be used only in `validate_gamma.py`; it is not imported or called by the production Gamma application.

Sources: [1], [2], [5].

Backup: Complete P7 requirements – D2 additions

ID	Status	Updated requirement	Verification
GG-GUI-01	New	The software shall provide a Tkinter GUI with an input field, Calculate control, Clear control, status label, and output/history area.	GUI inspection
GG-SC-01	New	The production code shall implement sine, exponential, logarithm, positive-base power, and absolute-value support as project-defined subordinate functions.	Code inspection
GG-SC-02	New	The production Gamma calculation shall not call Python's math-library Gamma, sine, exponential, logarithm, square-root, or power functions.	Code inspection
GG-BI-01	New	The production numerical calculation shall avoid <code>range()</code> and <code>len()</code> by using <code>while</code> loops and manual counters where needed.	Code inspection
GG-EX-01	New	Expected input, domain, range, and underflow cases shall be represented by user-facing exception classes.	Exception test
GG-DEP-01	New	The application shall run from a terminal command without depending on a specific IDE.	Terminal test

Separate P6 project constraints

Public repository, README, and meaningful commit messages are required project evidence for P6, not function requirements in P7.

Sources: [1], [2].

Backup: Complete validation table

x	Gamma Guard result	Validation reference	Relative error	Status
0.1	9.51350769866873	9.51350769866873	1.867×10^{-16}	Pass
0.5	1.77245385090552	1.77245385090552	3.758×10^{-16}	Pass
1	1	1	0	Pass
1.5	0.886226925452758	0.886226925452758	3.758×10^{-16}	Pass
2.5	1.32934038817914	1.32934038817914	3.341×10^{-16}	Pass
5	24	24	1.332×10^{-15}	Pass
-0.5	-3.54490770181103	-3.54490770181103	7.517×10^{-16}	Pass
-1.5	2.36327180120735	2.36327180120735	7.517×10^{-16}	Pass

Reference policy

`math.gamma()` is used only in the validation script. It is not called by the Gamma Guard implementation.

Sources: [3], [5].

References

- [1] D1: Gamma Guard, F4: $\Gamma(x)$, Md. Ariful Hossain, SOEN 6011, 2026.
- [2] SOEN 6011, "Project Description," Concordia University, Summer 2026.
- [3] NIST Digital Library of Mathematical Functions, "Gamma Function," ch. 5. [Online]. Available: <https://dlmf.nist.gov/5>.
- [4] R. Munafo, "Coefficients for the Lanczos Approximation to the Gamma Function," 2019. [Online]. Available: <https://www.mrob.com/pub/ries/lanczos-gamma.html>.
- [5] Python Software Foundation, "math – Mathematical functions," Python documentation. [Online]. Available: <https://docs.python.org/3/library/math.html>.
- [6] H. Tavakoli, *Prompt Engineering for Everyone: A Self-Taught, Human-Centered Approach to AI Programming*. Berkeley, CA, USA: Apress, 2026.
- [7] Python Software Foundation, "tkinter – Python interface to Tcl/Tk," Python documentation. [Online]. Available: <https://docs.python.org/3/library/tkinter.html>.
- [8] GitHub Docs, "About READMEs" and "About commits," GitHub documentation. [Online]. Available: <https://docs.github.com/>.